# Shantanu Maratha

ECE Undergraduate | Embedded Systems Engineer (ESP32, Sensors, IoT)

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**Linkedin** | **Portfolio** | **GitHub**


## Education

- **Thapar Institute of Engineering and Technology** — Patiala, Punjab
  Bachelor of Technology - Electronics and Communication Engineering — **August 2025 - Present**

## Skills Summary

- **Embedded Systems**: ESP32, Arduino, I2C, UART, SPI, MQTT, Real-time Processing
- **Programming**: C, C++ (Arduino), Python, JavaScript
- **Hardware / Electronics**: Sensors, Electromagnets, Power Electronics, GPS Modules, OLED Displays
- **Tools / Platforms**: Git, GitHub, KiCad, Arduino IDE, ESP-IDF, Vercel

## Projects / Open-Source

- **Zephyr-Station** — **ESP32, C++**
  Designed an ESP32-based environmental monitoring system integrating multiple sensors via I2C with OLED visualization, SD-card data logging, and threshold-based alert mechanisms.

- **Jolt-Locator** — **ESP32, C++**
  Developed a GPS-based true-north navigation system on ESP32 with real-time heading computation and location processing using UART communication.

- **The-Ruin-Machine** — **ESP32, C++**
  Implemented a probability-based stochastic simulator on ESP32 with physical I/O interfacing (OLED display and buzzer actuation).

- **Geometrical Shape Detection | Report** — **Python, OpenCV**
  Developed 6 image analysis algorithms using OpenCV and NumPy for automated geometrical shape detection and recognition with modular Python implementations.

## Experience

- **Ragastra MV Lev** — Patiala, Punjab
  Hardware & Controls Engineer — **September 2025 - Present**
  - **Electromagnetic System Design**: Designed and tuned custom electromagnets for magnetic levitation prototype with optimized coil parameters.
  - **Power Electronics**: Designed and debugged SMPS-based power circuitry for inductive electromagnetic loads.
  - **Motor Control Systems**: Implemented Arduino-based synchronized dual stepper motor control with real-time actuation.
- **Grosity** — Patiala, Punjab
  Early-Stage Founding Engineer — **October 2025 - February 2026**
  - **Web Development**: Developed and deployed company website using AI-assisted development workflows (Claude, GitHub Copilot, Kiro IDE).
  - **Autonomous Systems R&D**: Contributed to embedded electronics and control logic for autonomous payload-delivery drone systems.
  - **Product Strategy**: Aligned engineering decisions with B2B and D2C product models and business strategy.

## Ongoing Research

- **GHFCVs - Reverse Engineering Study** — **December 2025 - Present**
  Literature review and comparative analysis of DC-DC converter topologies with hands-on teardown of commercial GHFCV systems to understand power electronics architecture and operational principles.

## Certifications & Achievements

- **Internet of Things** - MIT Professional Education x Santander Open Academy
- **Google: Artificial Intelligence and Productivity** - Google x Santander Open Academy
- **National Finalist** - E Summit Ideastorm 2026, IIT Roorkee